2b. Content of Computer systems (J276/01)

This component will introduce learners to the Central Processing Unit (CPU), computer memory and storage, wired and wireless networks, network topologies, system security and system software.
It is expected that learners will become familiar with the impact of Computer Science in a global context through the study of the ethical, legal, cultural and environmental concerns associated with Computer Science.

Learners may draw on some of this content when completing the Programming Project.

1.1 Systems architecture

Learners should have studied the following:

• the purpose of the CPU
• Von Neumann architecture:
  ○ MAR (Memory Address Register)
  ○ MDR (Memory Data Register)
  ○ Program Counter
  ○ Accumulator
• common CPU components and their function:
  ○ ALU (Arithmetic Logic Unit)
  ○ CU (Control Unit)
  ○ Cache
• the function of the CPU as fetch and execute instructions stored in memory
• how common characteristics of CPUs affect their performance:
  ○ clock speed
  ○ cache size
  ○ number of cores
• embedded systems:
  ○ purpose of embedded systems
  ○ examples of embedded sytems.